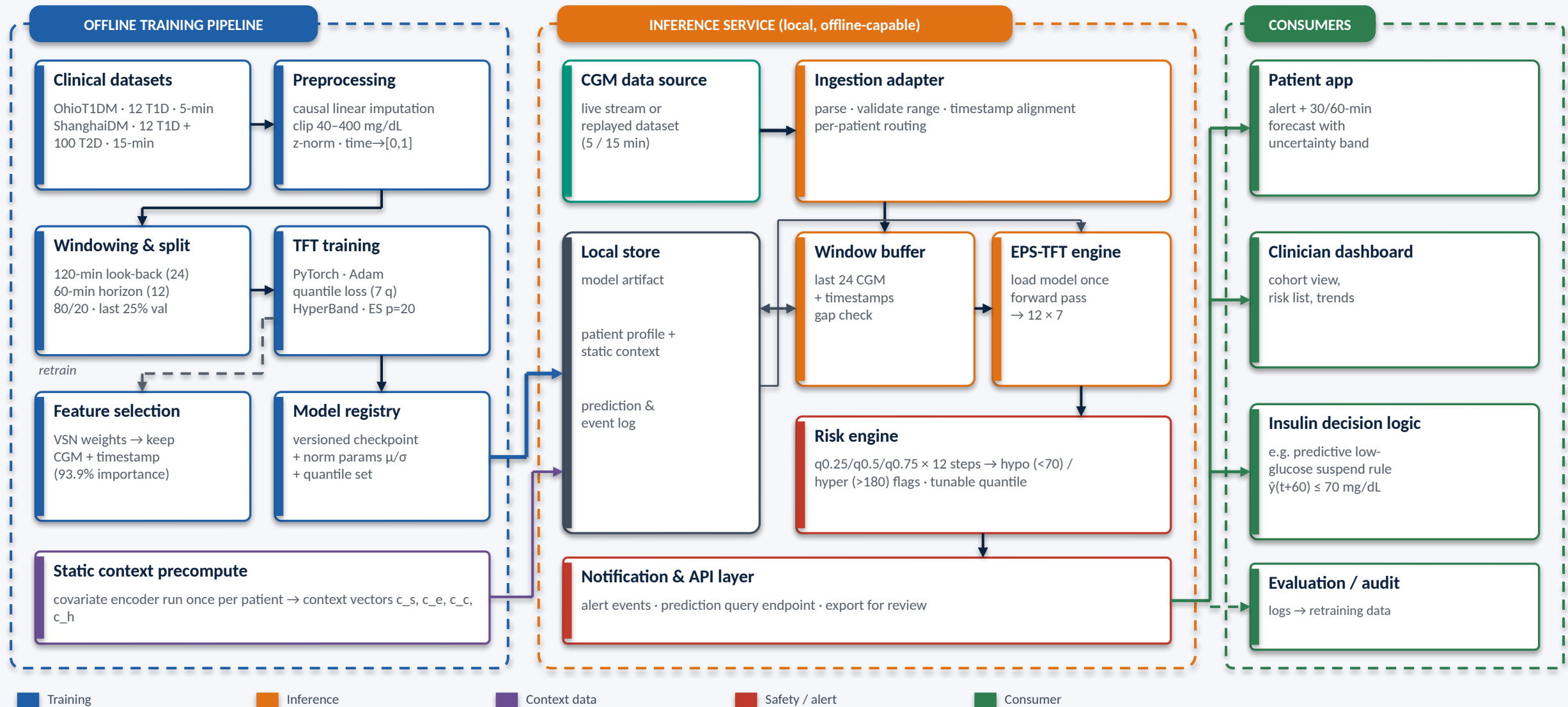


GlucorAG / EPS-TFT

Population-specific, multi-horizon blood-glucose forecasting with a
Temporal Fusion Transformer — data, model, inference and decision-support software

- 1 Horizontal end-to-end architecture (training pipeline → inference service → consumers)
- 2 Vertical layered architecture (6-layer software stack with interfaces)
- 3 TFT model architecture — tensor-level data flow
- 4 Software module & component architecture
- 5 Runtime sequence (one prediction cycle per CGM sample)
- 6 Data & model lifecycle (MLOps) pipeline
- 7 Non-functional requirements & design decisions

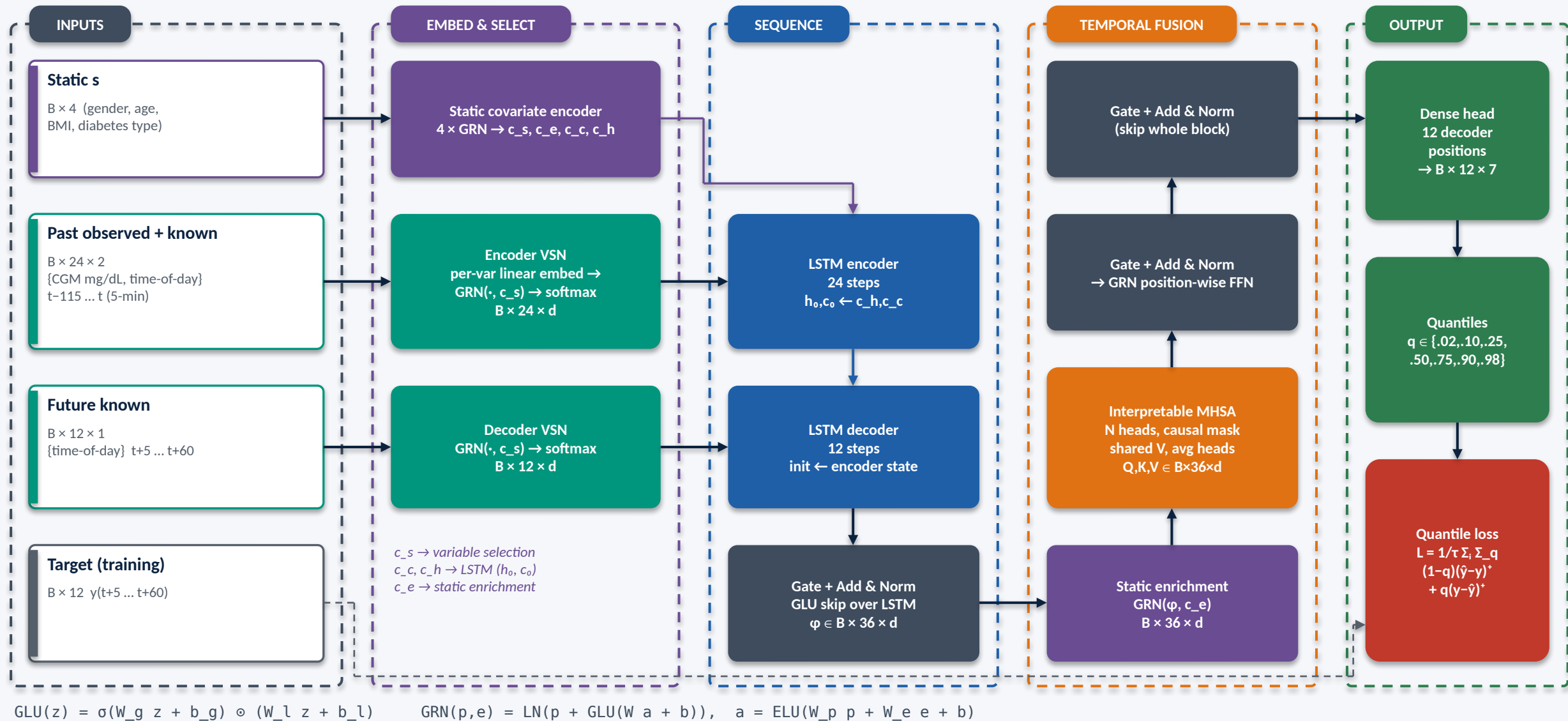
1 • Horizontal End-to-End Architecture



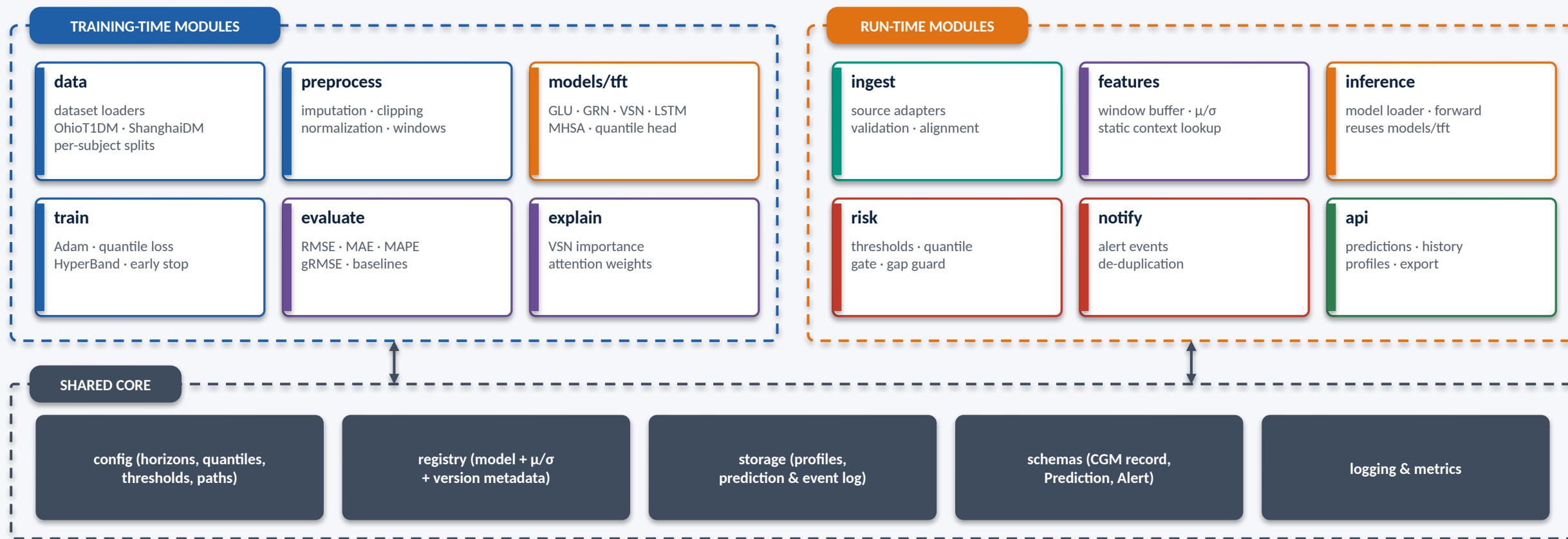
2 · Vertical Layered Architecture



3 · Temporal Fusion Transformer — Tensor-Level Architecture



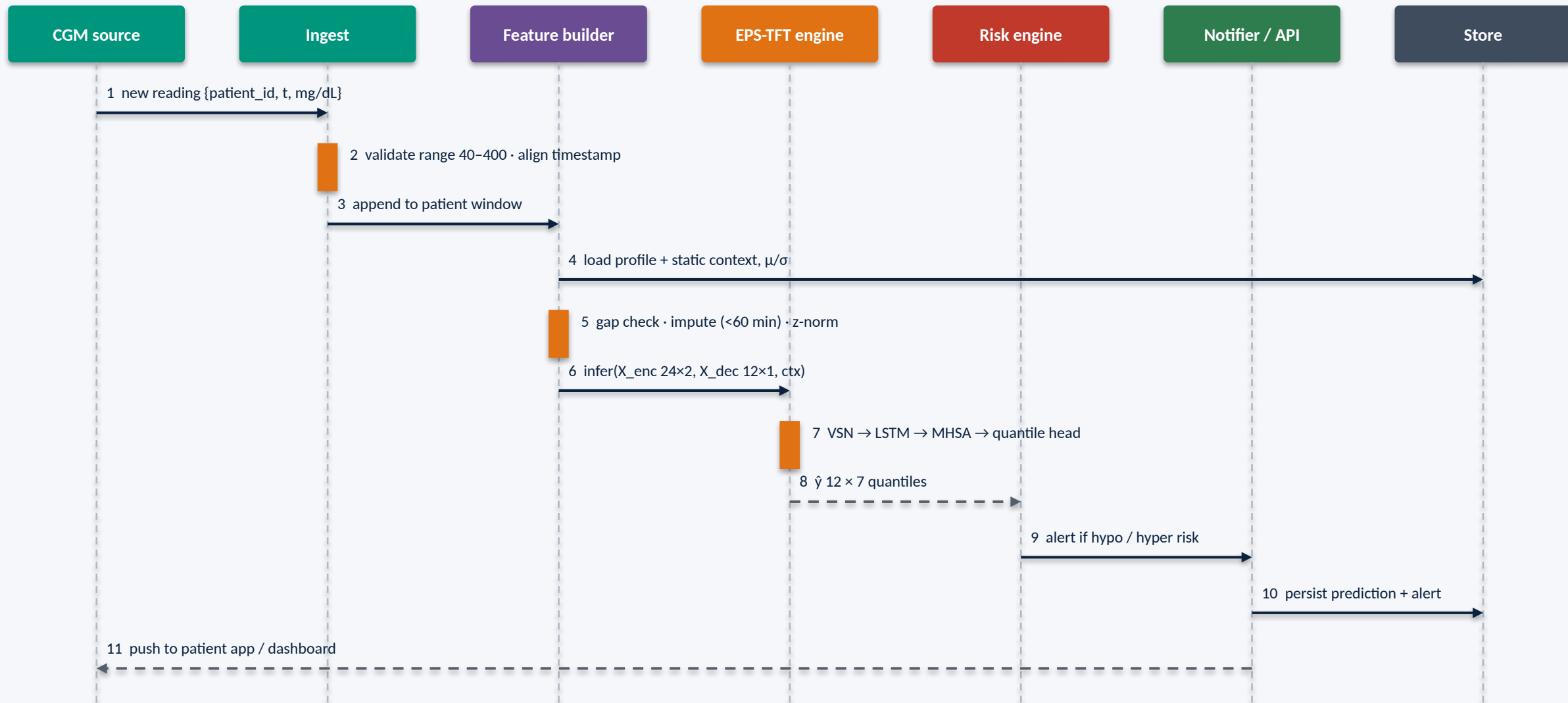
4 · Software Module & Component Architecture



Key data contracts

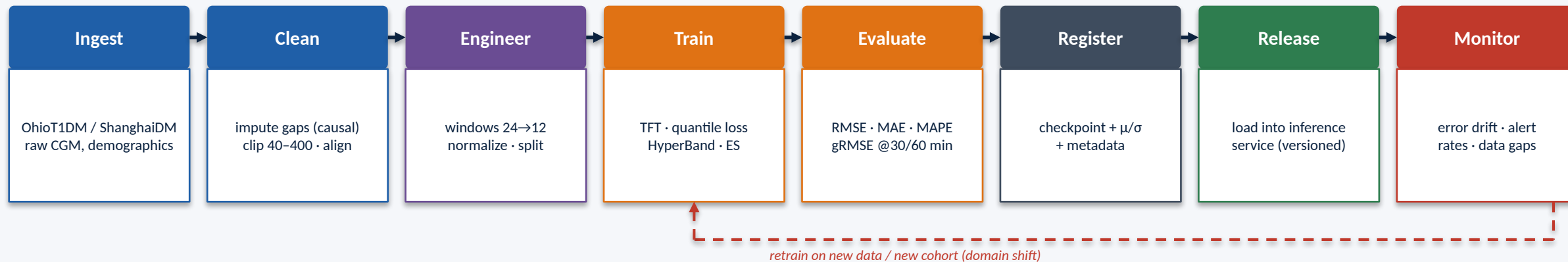
```
CGMRecord {patient_id, timestamp, glucose_mg_dL}    PatientProfile {patient_id, age, gender, bmi, diabetes_type, context_vectors}
Prediction {patient_id, t0, horizons[12], quantiles[7], values[12x7], model_version}    Alert {patient_id, type: hypo|hyper|data_gap, horizon_min, severity, t_raised}
```

5 • Runtime Sequence — One Prediction Cycle



If gap > 60 min at step 5: skip inference, raise data_gap alert. Budget: whole cycle must finish well within one CGM sampling period (5 min).

6 • Data & Model Lifecycle (MLOps) Pipeline



Validation gates

- Held-out test: last 20% per subject
- Paired t-test vs next-best ($p < 0.05$)
- 5-fold cross-individual CV (ShanghaiDM: 14.7 / 23.5 mg/dL RMSE)
- Ablation: covariate encoder, LSTM decoder
- Baselines: LR, SVR, XGBoost, LSTM, N-BEATS, N-HITS

Run-time safeguards

- Missing CGM: impute gaps < 60 min; longer → suspend prediction + alert
- No-reading watchdog → data_gap alert
- Quantile-based alarm sensitivity
- Prediction & event log for review
- Model version pinned per prediction

Artifacts & versioning

- Dataset snapshot + preprocessing config
- Trained checkpoint + hyperparameters
- Normalization μ/σ , quantile set
- Precomputed static context per patient
- Evaluation report per release
- Model card (scope, cohorts, limits)

7 · Non-Functional Requirements & Key Design Decisions

Latency	prediction ready well within one 5-min CGM period
Accuracy	lowest RMSE/MAE/MAPE/gRMSE vs 6 baselines
Generalization	one model for T1D + T2D cohort; cross-individual CV
Privacy	local inference; raw CGM not sent to a cloud
Availability	no internet dependency for predictions
Interpretability	VSN feature weights + attention weights
Safety	quantile bounds; gap guard; tunable alarms

D1 Population model, not per-patient

One TFT for the whole cohort; static demographics personalise via covariate encoder. Removing it raises RMSE 21.7 → 22.4 mg/dL.

D2 Direct multi-horizon output

All 12 steps in one pass: no recursive error accumulation; one model serves 30 & 60 min.

D3 Minimal inputs: CGM + time

VSN shows 93.9% importance; no manual meal/insulin logging, less compute.

D4 Precompute static context

Covariate encoder runs once per patient; inference reuses stored context vectors.

D5 Quantile (probabilistic) output

7 quantiles give an uncertainty band; alarm sensitivity is a config choice, not a retrain.

D6 Training and inference share one model package

Same TFT code in both paths avoids train/serve skew; versioned artifacts make results reproducible.